Murtaza S. Ahmed

Embedded & Distributed Systems Engineer (Distributed focus)

murtazaofficial@proton.me | +91 9390341984 | onerealti.github.io/portfolio | github.com/onerealti | linkedin.com/in/onerealti

Computer Engineering student specializing in embedded edge AI systems, distributed systems development, and backend infrastructure. Experienced in TensorRT-accelerated inference pipelines on NVIDIA Jetson Nano, decentralized media provenance protocols using IPFS/Arweave, and deterministic Linux system provisioning.

Professional Experience

Undergraduate Research Assistant (Edge AI Systems) — **Dec 2025 -- Present**
Department of Mechanical Engineering (MJCET) — *Hyderabad, TG*

- Developed an embedded edge AI platform using an NVIDIA Jetson Nano, IMX719 camera module, and telemetry sensors.
- Optimized computer vision inference pipelines using TensorRT FP16 acceleration, achieving 40-70 ms latency (14-20 FPS) under constrained compute environments.
- Implemented deterministic execution and monitoring systems for stable long-duration deployment.
- Researched adaptive workload scheduling and hardware-aware resource allocation strategies for heterogeneous LLM inference systems in interdepartmental R&D program.
- Exploring fault-tolerant orchestration approaches for distributed AI workloads across mixed compute environments.

Builder & Systems Architect — **July 2025 -- Present**
Verity (Decentralized Verification Platform) — *Hyderabad, TG*

- Designing and implementing system to verify digital media provenance using blockchain and decentralized storage.
- Integrating AI modules for manipulation detection and automated claim extraction.
- Developing media provenance infrastructure using distributed storage systems including IPFS and Arweave.
- Building concurrent ingestion pipelines and immutable audit logging mechanisms for ML-based verification workflows.

Projects

Smart Agri Four-Legged Bot (SAFL-B) (*C, Python, UART/Serial, I2C, GPIO, Hardware Control*) — **Dec 2025**

- Awarded 1st Place at the MAKEFORHYDERABAD 2-Day Makeathon for autonomous robotics systems integration.
- Implemented deterministic UART/Serial communication protocols for multi-joint motor control.
- Developed telemetry stabilization routines to reduce terrain-induced sensor and image noise.

Technical Skills

Languages:	C, Python, Bash, PostgreSQL, Git
Systems:	Linux, Docker, Kubernetes
ML/Edge Computing:	TensorRT, NVIDIA Jetson Nano
Hardware & Interfaces:	UART/Serial, I2C, GPIO

Education

Bachelor of Engineering in Mechanical Engineering — **May 2028**
Muffakham Jah College of Engineering & Technology (Osmania University) — *Hyderabad, TG*

- Relevant Coursework: Scientific Computing, Embedded Systems Programming, System Modeling & Simulation